

TITLE:

GENERATOR MAINTENANCE PROCEDURE

Revision No.	Brief Description of Change	Effectivity Date
0	Origination	

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GENERATOR MAINTENANCE PROCEDURE**1.0 PURPOSE & SCOPE**

Generators are the backbone of power continuity for businesses and various industries. A well-maintained generator ensures uninterrupted operations during outages, minimizing costly downtime. Regular commercial generator maintenance is essential to ensure reliability and longevity, helping businesses avoid emergency failures and unexpected expenses.

2.0 REFERENCES

- Quality manual
- www.merriam-webster.com
- www.collinsdictionary.com

3.0 DEFINITIONS & ABBREVIATIONS:**3.1 DEFINITIONS**

- **Generator maintenance** – Planned inspections, servicing, testing, and repairs performed to keep a generator set reliable, safe, and ready to supply power when required.
- **Preventive maintenance** – Scheduled routine tasks (inspections, oil/filter changes, battery checks, test runs) intended to prevent failures before they occur.
- **Corrective maintenance** – Repairs or replacements performed after a fault or failure to restore normal operation.
- **Predictive maintenance** – Condition-based actions driven by monitoring and diagnostics (vibration, oil analysis, fuel testing) to fix issues just before failure.
- **Automatic transfer switch ATS** – A device that automatically transfers electrical load between utility power and generator power during an outage.
- **Automatic voltage regulator AVR** – A control that maintains generator output voltage within specified limits.
- **Battery maintenance** – Inspection, cleaning, testing, and charging of starting batteries to ensure reliable engine starts.

3.2 ABBREVIATIONS

- **ATS** - Automatic Transfer Switch
- **AVR** - Automatic Voltage Regulator
- **CB** - Circuit Breaker
- **CT** - Current Transformer
- **LT** - Load Test
- **PM** - Preventive Maintenance
- **CM** - Corrective Maintenance
- **PdM** - Predictive Maintenance
- **OEM** - Original Equipment Manufacturer
- **NFPA** - National Fire Protection Association
- **AMF** - Automatic Mains Failure

4.0 GENERAL REQUIREMENTS:

- **Regular inspections:** Check oil, coolant, fuel, belts, hoses, and electrical connections.
- **Scheduled servicing:** Perform oil and filter changes, coolant replacement, and fuel system cleaning at manufacturer-specified intervals.
- **Battery care:** Test, clean, and charge starting batteries to ensure reliable engine starts.

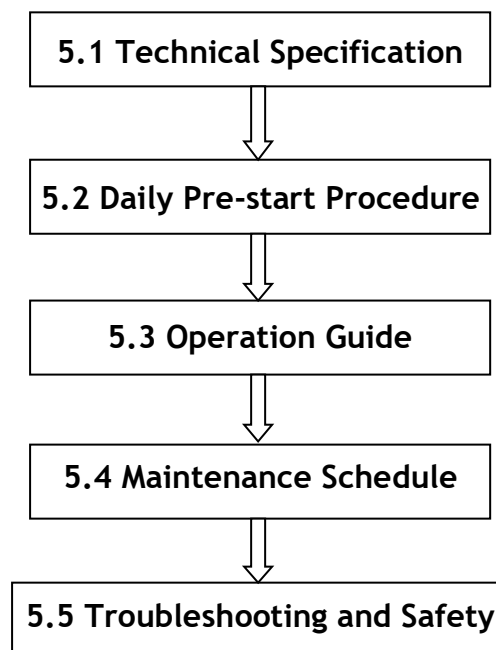
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- **Operational testing:** Run the generator monthly under load to verify performance and cooling.
- **Automatic Transfer Switch (ATS) checks:** Verify proper switching between utility and generator power.
- **Recordkeeping:** Maintain logs of inspections, test results, repairs, and parts replaced.

5.0 PROCEDURE INSTRUCTIONS

PROCEDURE OUTLINE



6.0 PROCEDURE DETAILS

Responsible Person	Activities	Interface Document/ Person										
	5.1 Technical Specification											
Generator Specialist, BM Supervisor & Technician	Understanding the physical and operational limits of the KTAA19-G6A is essential for preventing overloads or mechanical failure.	OEM Cummins KTAA19-G6A Service Manual										
	<table><tr><td>Parameter</td><td>Specification</td></tr><tr><td>Engine Model</td><td>Cummins KTAA19-G6A</td></tr><tr><td>Power Output</td><td>Prime: 545kW (681kVA) / Standby: 600kW (750kVA)</td></tr><tr><td>Rated Speed</td><td>1800 RPM (60Hz) or 1500 RPM (50Hz)</td></tr><tr><td>Lube Oil Capacity</td><td>50 Liters (13.2 Gallons)</td></tr></table>		Parameter	Specification	Engine Model	Cummins KTAA19-G6A	Power Output	Prime: 545kW (681kVA) / Standby: 600kW (750kVA)	Rated Speed	1800 RPM (60Hz) or 1500 RPM (50Hz)	Lube Oil Capacity	50 Liters (13.2 Gallons)
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Generator Specialist, BM Supervisor & Technician	Coolant Capacity	-116.5 Liters (Engine + Radiator)	OEM Cummins KTAA19-G6A Service Manual
	Fuel System	Cummins PT (Pressure-Time)	
	Electrical System	24V DC	
	Aspiration	Turbocharged & Air-to-Air Aftercooled	
5.2 Daily Pre-Start Procedure			
BM Supervisor and Technician	Think of this as your "flight check." Never start the engine without verifying these fluid levels and physical conditions. • Lube Oil: Pull the dipstick and wipe it clean. Re-insert and check that the level is between the 'L' (Low) and 'H' (High) marks. Use SAE 15W-40 (API CH-4 or higher). • Coolant: Check the radiator sight glass. The mixture must be a 50/50 ratio of water and ethylene glycol to prevent corrosion and boiling. • Fuel System: * Drain water/sediment from the Fuel-Water Separator. ○ Ensure the day tank is at least 75% full to prevent air from entering the lines. • Air Cleaner: Inspect the restriction indicator. If the red flag is visible, the filter is clogged and must be serviced before operation. • Battery: Check the control panel. At rest, the 24V system should read at least 25.2V.		Inspection Checklist
5.3 Operation Guide			
BM Supervisor, Technician	Whether you are using a Power Command® or Smartgen controller, the logic remains the same. Manual Starting Sequence 1. Isolate Load: Ensure the main output circuit breaker is in the OFF (Open) position. Never start a generator under load. 2. Power On: Switch the control panel to Manual Mode. 3. Crank: Press the Green (Start) button.		BM Supervisor, Technical, Checklist

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 Technician

4. Warm-up: Allow the engine to idle for 3-5 minutes.

- *Normal Oil Pressure:* 345-483 kPa (50-70 psi).
- *Idle Speed:* 675-775 RPM.

Loading the Generator

- Once the frequency (60Hz/50Hz) and voltage are stable, close the main circuit breaker.
- **Critical Note on "Wet Stacking":** Avoid running this engine at less than **30% load** for long periods.¹ Low-load operation leads to unburnt fuel in the exhaust, which can damage the cylinders and turbochargers.²

Controlled Shutdown

1. **Remove Load:** Open the main circuit breaker.
2. **Cool-down:** Let the engine run at no-load for **5 minutes**. This allows the turbochargers to cool down gradually. Stopping a hot engine immediately can "cook" the oil inside the turbo bearings.
3. **Stop:** Press the **Red (Stop)** button.

 BM Supervisor,
 Technical,
 Checklist

5.4 Maintenance Schedule

 BM Supervisor, Service
 Contractor

Staying ahead of the "Overhead Set" is the secret to making a K19 series engine last for decades.

Interval	Key Tasks
Daily	Check Oil, Coolant, and Fuel-Water separator.
250 Hrs / 6 Months	Change Lube Oil & Filters; Replace Fuel and Coolant filters.
1500 Hrs / 1 Year	Full Engine Clean; Adjust Overhead Set (Valves/Injectors); Check Zinc Anodes.
6000 Hrs / 2 Years	Flush Cooling System; Inspect Vibration Dampers and Turbochargers.

 Preventive
 Maintenance
 Checklist

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5.5 Troubleshooting & Safety		
BM Supervisor, Technician, Safety Officer	⚠ WARNING: The PT (Pressure-Time) Fuel System operates under extreme pressure. Never loosen or "crack" a fuel line while the engine is running. Low Oil Pressure Shutdown: If the unit shuts down for oil pressure, do not attempt a restart until you have manually verified the oil level. The KTAA19 can seize within seconds of a pressure loss. High Water Temp (>104°C): Check for debris blocking the radiator core or a loose fan belt. Engine Hunting/Surging: This is often caused by air bubbles in the fuel lines or a malfunctioning EFC (Electronic Fuel Control) governor.	Incident Report Form, Inspection Checklist, Safety Officer, OEM Technical Support
	Weekly / Monthly Inspection Checklist <i>(Check each box after verifying during the monthly run):</i> 1. Visual & Physical Checks ☐ Belts & Hoses: Check fan belt tension and heater hose flexibility. ☐ Exhaust System: Check for soot leaks at the manifold or silencer. ☐ Air Intake: Ensure the air cleaner restriction indicator is green. ☐ Oil Level: Check dipstick; keep level between L and H. ☐ Coolant Level: Check radiator/expansion tank level. ☐ Fuel/Water Separator: Drain any accumulated water from the primary filter bowl. ☐ General Leaks: Look for fresh oil, coolant, or fuel spots on the floor. ☐ Battery Charger: Verify the DC ammeter or "Charging" light is active. ☐ Fuel Supply: Ensure the main storage tank and day tank are at least 75% full. ☐ Enclosure: Ensure air intake louvers are clear of debris (leaves, paper, etc.). ☐ Battery Terminals: Check for white/green corrosion or loose cables.	

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 Technician, Safety
 Officer

- ☐ **Heaters:** Check the "Jacket Water Heater." The engine block should feel warm to the touch (critical for quick starting).
- ☐ **Control Panel:** Ensure no "Common Alarm" or "Fault" lights are active.

2. Operational Test (Under Load)

Note: Cummins recommends testing at a minimum of 30% load (180kW) to maintain engine health.

- ☐ **Warm-up:** Run for 5 minutes at no load.
- ☐ **Transfer:** Switch the Automatic Transfer Switch (ATS) or close the breaker to take load.
- ☐ **Load Duration:** Run under facility load for at least **30-60 minutes**.
- ☐ **Governor Performance:** Ensure speed (Hz) remains stable when load is applied.
- ☐ **Cool-down:** Run at no load for 5 minutes before total shutdown.

3. After-Test Status

- ☐ **Fuel Level:** Refill the day tank if necessary.
- ☐ **Control Panel:** Ensure the selector switch is back in "AUTO" mode.
- ☐ **Battery Charger:** Verify the "Float" light is on.

 Incident Report
 Form, Inspection
 Checklist, Safety
 Officer, OEM
 Technical Support

6.1 Appendices

- Preventive Maintenance Checklist
- Incident Report Form
- Purchase Requisition Form